6. $a = -3$, Remainder : 27 7. (i) $(x-1)$ is a factor (ii) $(x-1)$ is not a factor
8. $(x-5)$ is a factor of $p(x)$ 9. $m = 10$ 11. $k = 3$ 12. Yes

Exercise 3.4

1. (i) $x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6xz$ (ii) $p^2 + 4q^2 + 9r^2 - 4pq + 12qr - 6pr$
(iii) $8p^3 - 24p^2 - 14p + 60$ (iv) $27a^3 + 27a^2 - 18a - 8$

2.(i) 18,107,210 (ii) −32, −6, +90

3. (i) 14 (ii) $\frac{59}{70}$ (iii) 78 (iv) $\frac{78}{70}$

4. (i) $27a^3 - 64b^3 - 108a^2b + 144ab^2$ (ii) $x^3 + \frac{1}{y^3} + \frac{3x^2}{y} + \frac{3x}{y^2}$

5.(i) 941192 (ii) 1003003001

6. 29 7. 280 8. 335 9. 198 10. $\pm 5, \pm 110$

11. 36 12.(i) $8a^3 + 27b^3 + 64c^3 - 72abc$ (ii) $x^3 - 8y^3 + 27z^3 + 18xyz$

13.(i) −630 (ii) $\frac{-9}{4}$

14. 72xyz

Exercise 3.5

1.(i) $2a^2(1 + 2b + 4c)$ (ii) $(a-m)(b-c)$

2.(i) $(x+2)^2$ (ii) $3(a-4b)^2$

(iii) $x(x+2)(x-2)(x^2+4)$ (iv) $\left(m + \frac{1}{m} + 5\right)\left(m + \frac{1}{m} - 5\right)$

(v) $6(1+6x)(1-6x)$ (vi) $\left(a - \frac{1}{a} + 4\right)\left(a - \frac{1}{a} - 4\right)$

3. (i) $(2x + 3y + 5z)^2$

(ii) $(-5x + 2y + 3z)^2$ (or) $(5x - 2y - 3z)^2$

4. (i) $(2x + 5y)(4x^2 - 10xy + 25y^2)$

(ii) $(3x - 2y)(9x^2 + 6xy + 4y^2)$

(iii) $(a+2)(a-2)(a^2 + 4 - 2a)(a^2 + 4 + 2a)$

5. (i) $(x + 2y - 1)(x^2 + 4y^2 + 1 - 2xy + 2y + x)$

(ii) $(l - 2m - 3n)(l^2 + 4m^2 + 9n^2 + 2lm - 6mn + 3ln)$

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